

Fault History, Case Scenarios and Diagnostic Evidence

Stamping press 01 (press-stamp-01) | Main drive and lubrication circuit

Machine	Stamping press 01 (machine ID press-stamp-01, type press, area Press shop, site vit-lab), stamping press in a car plant
Sensors	MAIN_MOTOR_CURRENT (A), LUBE_OIL_PRESSURE (bar), BEARING_VIBRATION_RMS (mm/s) and the 64-bin spectrum BEARING_SPECTRUM (mm/s); sensor IDs PRESS-STAMP-01.<CHANNEL_KEY>
Components / subsystem	Press main drive and its pressure lubrication circuit: main drive motor, lubrication unit (gear pump, reservoir, lube filter) and main bearing housing; planned agent tags main_drive_motor, lube_filter, lubrication_unit, main_bearing_housing
Document title	Fault History, Case Scenarios and Diagnostic Evidence
Document ID	PS01-DOC-05
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Purpose	Provides case-based diagnostic knowledge for press-stamp-01: eighteen worked scenarios showing how telemetry of the main drive motor current, lube oil pressure and bearing vibration is interpreted, differentiated, repaired and verified, with comparison tables of the evidence.
Scope	Simulated and illustrative cases only. The repository holds no real fault history; cases C01 to C06 and C15 to C18 are built from the repository's two simulated faults [P][D], the others are illustrative [I]. Limits are in Document 01, signatures in Document 02, the procedure in Document 03, maintenance in Document 04.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_PRESS_STAMP_01.md (catalog limits, simulation equations, topics, runbook stubs, tests).
[D]	Derived value: computed from the repository's own simulation model (seed 1, 2 Hz, 600 s per case) or algebraically from its equations, as recorded in the context file.
[E]	Engineering reference: general industrial practice for press drives, lubrication circuits and vibration diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. There is no physical press or transducer. An ESP32 gateway (or a software simulator) plays the press controller and generates values from documented equations. The repository describes its limits as illustrative constants chosen for the demo, not vendor values, and its runbooks as not an OEM manual. Values tagged [P] are therefore prototype values, not OEM specifications; values tagged [E] and [I] are not manufacturer-certified.

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1. Fault History Status and Use of the Cases

1.1 Fault history

There are no recorded incidents, work orders, failure dates or repair records for press-stamp-01 in the repository, and stored telemetry expires after 48 h [P]. Every case in this document is a simulated or illustrative scenario. No case is a real plant event, and none may be cited as failure statistics.

Table 1. Basis of the cases

Basis	Tag	Cases	How the numbers were obtained
Repository simulation (CLOGGED_FILTER, BEARING_WEAR)	[P] / [D]	C01 to C06, C15 to C18	Equations and the 600 s derived tables of the context file; linear fits within about 0.05 mm/s for vibration means between tabulated r values
Illustrative scenarios of faults not modelled in the repository	[I]	C07 to C14	Qualitative signatures from engineering practice with illustrative values, labelled in each case

1.2 Reproducing the project cases

The simulator is seeded and can be accelerated. Cases C01 to C06 can be re-created with node server/scripts/simulate-devices.mjs --machines press-stamp-01 --scenario CLOGGED_FILTER --ramp 120 (or BEARING_WEAR), with --seed and --speed as needed, or from the Faults tab (needs ENABLE_DEVICE_COMMANDS true) or cmd/scenario [P]. The scenario and ramp are published only in diag.sim and served only by GET /devices/:machineId/sim; the case reasoning below uses channel evidence only [P].

1.3 Case record format

Table 2. The 16 elements of every case

#	Element	Content
1	Case ID	C01 to C18
2	Machine state	Operating and gateway state
3	Sensor readings and pattern	Values and shape
4	Baseline	Healthy reference
5	Threshold comparison	Level against Document 01
6	Symptoms	What an operator would notice
7	Candidate faults	From Document 03 step 8
8	Evidence supporting	For each candidate
9	Evidence against alternatives	Why others are rejected
10	Diagnostic reasoning	Rules applied (R-xx)
11	Final diagnosis	Fault ID, confidence tier
12	Recommended inspection	Document 04 task
13	Corrective action	Action
14	Expected post-maintenance telemetry	Values
15	Verification	Criteria
16	Basis	Tag and source

2. Case Summary Table

Table 3. Case summary

Case	Situation type	Fault	Key signature	Final diagnosis	Level	Basis
C01	Gradual degradation	CLOGGED_FILTER, 120 s ramp	P60 falls, V60 rises, flat floor	F01	L1 to L3	[P][D]
C02	Rapid, severe	CLOGGED_FILTER, 60 s ramp	Alarm on P and V within 45 s	F01	L3	[P]
C03	Early-stage fault	BEARING_WEAR r 0.4	V60 1.94 mm/s, lines, P normal	F02	L1	[P][D]
C04	Sustained threshold violation	BEARING_WEAR r 1.0	V60 2.89 mm/s above warn, no alarm	F02	L2	[P][D]
C05	Intermittent abnormality	BEARING_WEAR r 0.7	V flickers across 2.5 mm/s	F02	L1 to L2	[P][D]
C06	Severe / critical	CLOGGED_FILTER r 1.0	P 2.47 bar, V 4.98 mm/s	F01	L3 to L4	[P][D]
C07	Mixed evidence	Clog plus bearing (not modelled)	Falling P, floor and lines	F16 (ambiguous with F01)	L2	[I]
C08	Sensor issue	Pressure transducer stuck	P flat 3.2 bar, no ripple, V normal	F12	LS	[I]
C09	Sensor issue	Vibration mount loose	V spikes, RMS differs from spectrum	F13	LS	[I]
C10	Sustained current warning	Die overload	lpk 182 A, idle 88 A	F09	L2	[I]
C11	False alarm / transient	None	Single spike, window mean normal	None	L0	[I]
C12	Opposite direction	Blockage after transducer	P rises	F05	Other	[I]
C13	Unresolved evidence	Pump wear	P falling with changed ripple, V not yet up	F12 or F03 (unresolved)	L2	[I]
C14	Data state	Stale then reboot	No telemetry, bootId changes	F15	LS	[P]/[I]
C15	Similar telemetry, different faults	Clog r 0.43 and bearing r 1.0	V60 about 2.9 mm/s in both	F01 and F02	L2	[P][D]
C16	Post-repair verification	Filter replaced	P and V return to baseline	Verification pass and a fail variant	L0	[P]
C17	Slow degradation	CLOGGED_FILTER, 600 s ramp	Drift rules lead; limits come late	F01	L1 to L3	[P][D]
C18	Threshold alarm baseline fails	BEARING_WEAR, 120 s ramp	Feature rule leads; alarm never fires	F02	L1 to L2	[P][D]

3. Detailed Cases: Project-Based

3.1 C01 Gradual clogged filter at the default 120 s ramp

Table 4. C01: case record

Element	Description
Case ID	C01
Machine state	Running at about 13 strokes per minute; status online, no gaps; fault ramp rises linearly over 120 s
Sensor readings and pattern	P60 falls steadily from 4.50 bar; V60 rises from 1.31 mm/s; HB300 grows from 0.03 mm/s; I60 rises about 3 percent over the ramp. Snapshots in Table below
Baseline	P60 4.50 bar, V60 1.31 mm/s, I60 94.8 A, HB300 0.030 mm/s
Threshold comparison	V60 passes Watch (1.50) at about 6 s, platform samples reach warn from about 30 s, V60 reaches 2.5 mm/s at about 40 s, P60 reaches 3.6 bar at about 53 s, both reach alarm at about 89 s
Symptoms	Operator may notice a low gauge reading and bearing noise from about 60 s [E]
Candidate faults	F01 CLOGGED_FILTER, F16, F03, F04, F12, F02
Evidence supporting each candidate	F01: P60 falls, HB300 above 0.24 mm/s with PMR300 below 2.5, delta V / delta P about -1.8 mm/s per bar, delta I / delta V about 0.8 A per mm/s
Evidence against alternatives	F02 rejected (pressure not normal, no discrete lines); F12 rejected (V60 and HB300 confirm the fall); F03 and F04 less likely (pump ripple 0.12 bar present, floor flat and broadband)
Diagnostic reasoning	R-16 at about 12 s, R-17 at about 24 s, R-09 at 40 s, R-06 at 53 s; R-25 confirms; R-20 not needed because the evidence is not mixed
Final diagnosis	F01 CLOGGED_FILTER, High confidence (score 11, margin 5 at $r = 0.5$)
Recommended inspection	T1: filter cartridge; check media grade and for metal particles
Corrective action	Replace the cartridge (a short task, in the order of minutes) [P]; lock-out and tag-out; relieve line pressure
Expected post-maintenance telemetry	P returns to about 4.5 bar with 0.12 bar ripple; V settles to about 1.3 mm/s; HB300 about 0.03 mm/s [P]
Verification	P60 4.35 to 4.65 bar, V60 1.20 to 1.45 mm/s, HB300 at most 0.06 mm/s, 60 minutes without L1 or above
Basis	[P][D]: equations and derived tables; times are $r \times 120$ s

Table 5. C01 snapshots [D]

Time	r	P60 (bar)	V60 (mm/s)	I60 (A)	Level
0 s	0.00	4.50	1.31	94.8	L0
12 s	0.10	4.30	1.68	95.1	L1
24 s	0.20	4.09	2.05	95.4	L1
36 s	0.30	3.89	2.41	95.7	L1
48 s	0.40	3.69	2.78	95.9	L2
60 s	0.50	3.49	3.15	96.2	L2
72 s	0.60	3.29	3.52	96.5	L2
89 s	0.74	3.00	4.04	96.9	L3
120 s	1.00	2.48	4.99	97.6	L3

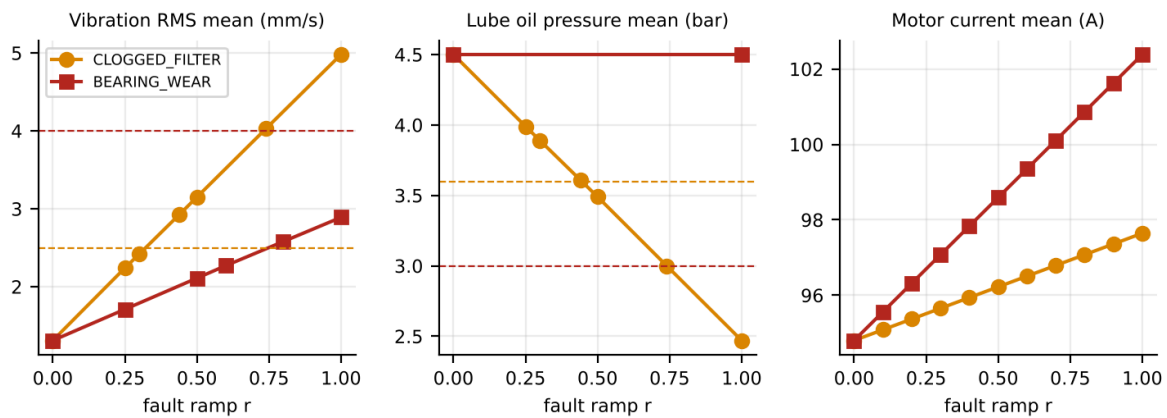


Figure 1. Progression of vibration, pressure and current means for the two simulated faults (derived table [D]); C01, C03 to C06 and C15 are points on these curves.

3.2 C02 Rapid severe clog (demo-script timeline, 60 s ramp)

Table 6. C02: case record

Element	Description
Case ID	C02
Machine state	Running; status online; ramp 60 s (demo script)
Sensor readings and pattern	About 20 s: vibration peaks cross warn and a broadband floor grows above 300 Hz; about 27 s: pressure under 3.6 bar; about 45 s: pressure under 3.0 bar and vibration over 4.0 mm/s [P]
Baseline	P60 4.50 bar, V60 1.31 mm/s
Threshold comparison	At 45 s (r about 0.75): P60 about 2.98 bar, V60 about 4.07 mm/s, I60 about 96.9 A: both alarm limits reached [D]
Symptoms	Rapid drop in pressure; noise; twin shows amber then red meshes on the lubrication unit and the main bearing housing [P]
Candidate faults	F01, F16, F03, F04
Evidence supporting each candidate	F01: pressure and vibration move together; floor is flat broadband; dl/dV about 0.8
Evidence against alternatives	F02 rejected (pressure falls); F03 and F04 cannot be excluded by telemetry alone but are less consistent with the flat floor
Diagnostic reasoning	R-09 at 20 s, R-06 at 27 s, R-10 and R-07 at 45 s, R-29 (time to alarm under 15 minutes) escalates +1 and R-30 (two channels) +1: L4 by escalation cap
Final diagnosis	F01 CLOGGED_FILTER, High confidence (score 13, margin 6)
Recommended inspection	T1 immediately; check bearing after
Corrective action	Replace cartridge now; after verification inspect the main bearings because the starved period was short but severe [P]
Expected post-maintenance telemetry	P returns to about 4.5 bar; V returns toward 1.3 mm/s; if V stays above 1.45 mm/s, treat as bearing damage (T2)
Verification	Section 14 criteria; extra spectrum check for lines
Basis	[P] demo script timing; [D] snapshot

3.3 C03 Early-stage bearing wear below every limit

Table 7. C03: case record

Element	Description
Case ID	C03
Machine state	Running; data valid
Sensor readings and pattern	P60 4.50 bar (unchanged); V60 1.94 mm/s; I60 97.8 A; discrete lines at about 88, 175, 263, 351, 439 Hz with sidebands; PMR300 above 3
Baseline	V60 1.31 mm/s, I60 94.8 A, no lines
Threshold comparison	V60 above Watch (1.50) and below warn (2.5); no platform warning; I60 below 99.5 A (Watch)
Symptoms	Slight increase in bearing noise [E]
Candidate faults	F02, F10, F11, F13, F01 early
Evidence supporting each candidate	F02: lines at defect orders (non-integer 3.55 to 17.75 x 1X), $\Delta I / \Delta V$ about 4.8 A per mm/s, P normal
Evidence against alternatives	F01 rejected (P60 4.50 bar, no floor); F10 and F11 rejected (no integer-order pattern); F13 rejected (RMS equals spectrum and current moves with V)
Diagnostic reasoning	R-15 (Watch), R-18, R-19, R-21 (drift), R-24
Final diagnosis	F02 BEARING_WEAR early, High confidence (score 11, margin 8)
Recommended inspection	T2: handheld analyser at the next planned stop
Corrective action	Plan a bearing replacement; do not run to alarm [P]; no urgent stop
Expected post-maintenance telemetry	After replacement V60 about 1.3 mm/s, lines absent, I60 about 94.8 A
Verification	Section 14 criteria
Basis	[P][D]: r about 0.4; $V60 = 1.31 + 1.58 r$

3.4 C04 Bearing wear at full severity: sustained warning, no alarm

Table 8. C04: case record

Element	Description
Case ID	C04
Machine state	Running; data valid
Sensor readings and pattern	V60 2.89 mm/s (range 2.68 to 3.21), 100 percent of samples at or over 2.5 mm/s; P60 4.50 bar; I60 102.3 A (max 164.1 A); strong lines at the defect orders [P][D]
Baseline	V60 1.31 mm/s, P 4.50 bar, I60 94.8 A
Threshold comparison	V above warn (2.5) continuously; below alarm (4.0) always; current below warn (175 A); pressure normal
Symptoms	Clear bearing noise and heat [E]
Candidate faults	F02, F16, F10, F11
Evidence supporting each candidate	F02: lines in all five bins at least 4 x healthy, PMR300 above 3, pressure normal, dI/dV 4.8
Evidence against alternatives	F01 rejected (P normal, no floor); F16 less likely (pressure normal)
Diagnostic reasoning	R-09 (L2), R-18, R-19, R-24; R-30 not applicable (one channel); the platform shows warn only, never alarm [P]
Final diagnosis	F02 BEARING_WEAR, High confidence (score 11, margin 8)
Recommended inspection	T2 immediately, T3 planning
Corrective action	Plan the replacement in the shortest planned stop; a planned change costs hours, an in-service failure costs days [P]
Expected post-maintenance telemetry	V60 about 1.3 mm/s; lines absent; I60 about 94.8 A
Verification	Section 14
Basis	[P][D]: $r = 1$

3.5 C05 Intermittent abnormality: vibration flickering across the warning limit

Table 9. C05: case record

Element	Description
Case ID	C05
Machine state	Running; data valid
Sensor readings and pattern	V60 about 2.42 mm/s (r about 0.7); samples flicker across 2.5 mm/s (2 percent of samples at r = 0.6, 88 percent at r = 0.8); lines present; P normal [P][D]
Baseline	V60 1.31 mm/s
Threshold comparison	Platform status alternates normal and warn per sample; V10 mostly below 2.5; PF60 between 2 and 88 percent
Symptoms	Intermittent amber tint on the main bearing housing
Candidate faults	F02, F13, F10
Evidence supporting each candidate	F02: lines, P normal, drift of V60 over hours
Evidence against alternatives	F13 rejected (RMS equals spectrum); F10 rejected (no integer pattern)
Diagnostic reasoning	R-35 not used (not a single spike), R-21 (drift), R-15, R-18; hysteresis ensures one opened investigation instead of repeated alerts
Final diagnosis	F02 BEARING_WEAR, High confidence (score 11, margin 8)
Recommended inspection	T2
Corrective action	Plan replacement at the next stop
Expected post-maintenance telemetry	As C03
Verification	Section 14
Basis	[P][D]: interpolated between tabulated r values; flicker share is a range, not a point value

3.6 C06 Severe clog at full severity

Table 10. C06: case record

Element	Description
Case ID	C06
Machine state	Running; data valid; ramp held at r = 1
Sensor readings and pattern	P60 2.47 bar (2.29 to 2.67); V60 4.98 mm/s (4.58 to 5.45), 100 percent in alarm; HB300 0.749 mm/s; I60 about 97.6 A [P][D]
Baseline	P60 4.50 bar, V60 1.31 mm/s
Threshold comparison	P at alarm (3.0) and below the proposed critical (2.5) by the window mean; V above alarm (4.0) and below the proposed critical 7.0 mm/s
Symptoms	Heavy noise; the twin shows red and pulsing marks on both meshes [P]
Candidate faults	F01, F16, F03, F04
Evidence supporting each candidate	F01: flat floor, P and V move together, delta V / delta P -1.8, I up about 3 percent
Evidence against alternatives	F02 impossible (bearing wear peaks at 3.2 mm/s and leaves P normal)
Diagnostic reasoning	R-07, R-08 (P10 at or below 2.5 bar), R-10; escalation to L4 [I]
Final diagnosis	F01 CLOGGED_FILTER, High confidence (score 13, margin 6)
Recommended inspection	T1 immediately, then T2 to check the starved bearings
Corrective action	Controlled stop; replace cartridge; inspect bearings for damage
Expected post-maintenance telemetry	P about 4.5 bar; V about 1.3 mm/s; if V stays high, bearing damage
Verification	Section 14 with an extended 60 minute run
Basis	[P][D]

4. Detailed Cases: Illustrative and Mixed

Cases C07 to C14 describe faults or conditions that the repository does not model. All values are illustrative [I]; signatures follow engineering practice. They show the reasoning, not recorded behaviour.

4.1 C07 Mixed evidence: lubrication first

Table 11. C07: case record

Element	Description
Case ID	C07
Machine state	Running; data valid
Sensor readings and pattern	P60 3.9 bar and falling; V60 3.2 mm/s; HB300 0.35 mm/s; PMR300 2.8; some discrete lines above a raised floor [I]
Baseline	P60 4.50 bar, V60 1.31 mm/s, HB300 0.03 mm/s
Threshold comparison	P at L1 (3.9 bar, above 3.6); V at L2 (3.2 mm/s)
Symptoms	Bearing noise; low pressure reading
Candidate faults	F16, F01, F02
Evidence supporting each candidate	F16: falling pressure and lines together; F01: falling pressure, raised floor; F02: lines
Evidence against alternatives	F02 less likely (pressure not normal); pure F01 unlikely because lines exist and PMR300 is between 2.5 and 3
Diagnostic reasoning	R-20: replace the filter first, then re-check the spectrum
Final diagnosis	F16 or F01, Medium confidence (F16 score 8, F01 score 5, margin 3: ambiguous)
Recommended inspection	T1 first, then T2 if lines remain
Corrective action	Replace the filter; if lines persist after the verification run, plan a bearing replacement
Expected post-maintenance telemetry	After the filter swap P about 4.5 bar and floor gone; lines remain if the bearing is worn
Verification	Section 14 plus a spectrum check for defect bins
Basis	[I]; combined fault not modelled in the repository

4.2 C08 Pressure transducer stuck

Table 12. C08: case record

Element	Description
Case ID	C08
Machine state	Running; data valid
Sensor readings and pattern	P60 3.20 bar with standard deviation near zero for 5 minutes; no 0.12 bar ripple; V60 1.30 mm/s; HB300 0.03 mm/s; I60 94.8 A [I]
Baseline	P60 4.50 bar with ripple 0.12 bar and noise 0.03 bar
Threshold comparison	P10 at or below 3.6 bar for 10 s: raw L2, but the signal is implausible: LS
Symptoms	Pressure alarm without any other symptom
Candidate faults	F12, F03, F04, F01
Evidence supporting each candidate	F12: flat signal, no ripple, no V or HB300 response
Evidence against alternatives	F01 rejected (V60 and HB300 normal); F03 and F04 less likely (no V response, pulsation absent rather than changed)
Diagnostic reasoning	R-03, R-34; sensor first
Final diagnosis	F12 pressure transducer fault, Medium confidence (score 7, margin 5)
Recommended inspection	T4: reference gauge at the same port
Corrective action	Replace or recalibrate the transducer; do not touch the filter first
Expected post-maintenance telemetry	P60 about 4.5 bar with ripple and noise; no change in V
Verification	P ripple visible; agrees with the reference
Basis	[I]

4.3 C09 Vibration sensor mount loosened

Table 13. C09: case record

Element	Description
Case ID	C09
Machine state	Running; data valid
Sensor readings and pattern	V10 jumps to 8 mm/s and back in steps; RMS differs from the root-sum-square of the spectrum bins by more than 2 percent; P60 4.50 bar; I60 94.8 A; spectrum shows spikes [I]
Baseline	V60 1.31 mm/s
Threshold comparison	V10 at or above 7.0 mm/s: raw L4, but contradicted by P and I: LS
Symptoms	Spurious vibration alarms
Candidate faults	F13, F02, F11
Evidence supporting each candidate	F13: RMS and spectrum disagree; step changes not seen in I
Evidence against alternatives	F02 rejected (no lines at defect orders; current does not move 4.8 A per mm/s); F11 rejected (no integer harmonic series)
Diagnostic reasoning	R-04, R-11 with contradiction, R-03; step-change evidence E12
Final diagnosis	F13 vibration sensor or mounting fault, Medium confidence (score 8, margin 6)
Recommended inspection	T5: mount, cable, handheld comparison
Corrective action	Refit or replace the cable
Expected post-maintenance telemetry	V60 about 1.3 mm/s; RMS equals spectrum
Verification	Handheld agrees; RMSchk below 2 percent
Basis	[I]

4.4 C10 Sustained current warning from process load

Table 14. C10: case record

Element	Description
Case ID	C10
Machine state	Running; data valid
Sensor readings and pattern	Ipk 182 A in every stroke; idle level about 88 A; I60 108 A; V60 1.35 mm/s; P60 4.50 bar [I]
Baseline	Ipk about 150 A, I60 94.8 A
Threshold comparison	Ipk at or above 175 A (L2); I60 above Watch; V and P normal
Symptoms	Heavier stroke sound
Candidate faults	F09, F08, F14
Evidence supporting each candidate	F09: increase confined to the working stroke; idle level unchanged; V and P unchanged
Evidence against alternatives	F08 rejected (idle level unchanged, no line-frequency lines); F14 rejected (press cycling, pulse present)
Diagnostic reasoning	R-12, R-23, R-26 (three strokes)
Final diagnosis	F09 process or die overload, Low confidence by score (5, margin 4) but clear classification
Recommended inspection	T8: die, material, shut height
Corrective action	Correct setup or material
Expected post-maintenance telemetry	Ipk about 150 A
Verification	Ipk and idle compared with baseline
Basis	[I]

4.5 C11 False alarm: single spike

Table 15. C11: case record

Element	Description
Case ID	C11
Machine state	Running; data valid
Sensor readings and pattern	One sample of BEARING_VIBRATION_RMS at 3.1 mm/s; the next samples 1.3 mm/s; V10 and V60 inside the healthy band [I]
Baseline	V60 1.31 mm/s; healthy 600 s maximum 1.72 mm/s [D]
Threshold comparison	Platform sample status alarm-free (3.1 is warn but below 4.0); V10 inside the limit
Symptoms	None
Candidate faults	Spike or data error; F13
Evidence supporting each candidate	No second sample, no change in P or I, spectrum unchanged
Evidence against alternatives	F02 and F01 rejected (no persistence, no trend)
Diagnostic reasoning	R-35, R-28: not persistent
Final diagnosis	No fault. Classified as a transient, not scored
Recommended inspection	No inspection; log the event
Corrective action	None
Expected post-maintenance telemetry	Features unchanged
Verification	L0 maintained
Basis	[I]

4.6 C12 Pressure rising: a different fault

Table 16. C12: case record

Element	Description
Case ID	C12
Machine state	Running; data valid
Sensor readings and pattern	P60 rises from 4.5 to 5.4 bar over minutes; V60 rises slowly from 1.3 to 1.6 mm/s; I60 unchanged [I]
Baseline	P60 4.50 bar
Threshold comparison	No project high limit; above the proposed 5.0 bar watch (I)
Symptoms	Bearing noise despite apparently good pressure
Candidate faults	F05, F06, F07, F12
Evidence supporting each candidate	F05: P rises, V rises later (less oil reaches the bearings)
Evidence against alternatives	F01 rejected (pressure rises, not falls: the repository says a rising reading points to a blockage after the transducer, a different fault [P]); F12 less likely (V also moves)
Diagnostic reasoning	R-22
Final diagnosis	F05 blocked nozzle or line after the transducer (not modelled); outside the evidence-weight matrix
Recommended inspection	Nozzles and lines after the transducer; relief valve
Corrective action	Clear the restriction
Expected post-maintenance telemetry	P about 4.5 bar; V about 1.3 mm/s
Verification	Section 14
Basis	[P] states the direction; values [I]

4.7 C13 Falling pressure with unresolved evidence

Table 17. C13: case record

Element	Description
Case ID	C13
Machine state	Running; data valid
Sensor readings and pattern	P60 3.2 bar and falling; pulsation amplitude changed and irregular; V60 1.4 mm/s; HB300 0.03 mm/s; I60 94.8 A [I]
Baseline	P60 4.50 bar
Threshold comparison	P60 at L2 (below 3.6 bar); V60 at baseline
Symptoms	Low pressure reading only
Candidate faults	F03, F04, F12, F01
Evidence supporting each candidate	F03 or F04: pressure falls with changed ripple; F12: pressure fall with no V response
Evidence against alternatives	F01 rejected (V60 and HB300 do not follow the fall)
Diagnostic reasoning	R-06 and R-34; evidence scores E1 and E14
Final diagnosis	F12 or F03/F04: unresolved (scores 4 and 5, margin 1)
Recommended inspection	T4 first (reference gauge), then T7 (level, leaks, pump)
Corrective action	No corrective action until the inspection resolves the ambiguity
Expected post-maintenance telemetry	Depends on the finding
Verification	Section 14
Basis	[I]

4.8 C14 Stale device and gateway reboot

Table 18. C14: case record

Element	Description
Case ID	C14
Machine state	status online then telemetry stops; after a few seconds a new bootId appears with seq back at 0
Sensor readings and pattern	No telemetry for more than 5 s (the stale threshold at the default 500 ms interval); then normal values; a simulated fault, if any, is cleared by the reboot [P]
Baseline	Healthy baseline
Threshold comparison	Not applicable: data state
Symptoms	The twin shows nothing or stale
Candidate faults	F15, F12 to F14
Evidence supporting each candidate	diag fields: uptimeS reset, reset reason, mqttReconnects
Evidence against alternatives	Mechanical faults cannot be judged without data
Diagnostic reasoning	R-01, R-02
Final diagnosis	F15 data-path or gateway event
Recommended inspection	Gateway diag, network
Corrective action	Fix the link or credential; recompute the baseline
Expected post-maintenance telemetry	status online, no gaps
Verification	Data valid for 60 minutes
Basis	[P] stale and reboot behaviour; scenario [I]

5. Similar Telemetry, Different Faults and Verification

5.1 C15 Vibration near 2.9 mm/s from two different faults

Table 19. C15: case record

Element	Description
Case ID	C15
Machine state	Two machines or two times, both running; data valid
Sensor readings and pattern	Case 15A: V60 2.89 mm/s, P60 3.63 bar (clog r about 0.43), I60 96.0 A, HB300 about 0.4 mm/s, flat floor. Case 15B: V60 2.89 mm/s, P60 4.50 bar (bearing r = 1), I60 102.3 A, HB300 low, lines [P][D]
Baseline	V60 1.31 mm/s, P60 4.50 bar
Threshold comparison	Both V60 above warn (2.5) and below alarm (4.0); 15A: P near warn; 15B: P normal
Symptoms	Identical vibration warning
Candidate faults	F01, F02, F16
Evidence supporting each candidate	15A: P60 falls and floor flat; 15B: P normal and lines
Evidence against alternatives	15A rejects F02 (no lines, P falling); 15B rejects F01 (P normal, no floor)
Diagnostic reasoning	R-09 then R-17 versus R-18 and R-19
Final diagnosis	15A: F01 (score 11, margin 5, High). 15B: F02 (score 11, margin 8, High)
Recommended inspection	15A: T1; 15B: T2
Corrective action	15A: replace cartridge; 15B: plan bearing replacement
Expected post-maintenance telemetry	15A: P 4.5 bar, V 1.3; 15B: lines absent, V 1.3
Verification	Section 14
Basis	[P][D]: demonstrates why the pressure and the spectrum shape matter more than the RMS value

5.2 C16 Post-repair verification: pass and fail variants

Table 20. C16: case record

Element	Description
Case ID	C16
Machine state	After T1 on the clog of C01 and after a run of 60 minutes
Sensor readings and pattern	Pass variant: P60 4.50 bar, V60 1.31 mm/s, HB300 0.03 mm/s, I60 94.8 A. Fail variant: P60 4.50 bar, V60 1.9 mm/s, lines present, I60 99 A [P][D]
Baseline	Pre-repair baseline recorded in the work order
Threshold comparison	Pass: L0. Fail: V60 at L1 (watch) with defect lines
Symptoms	Pass: none. Fail: residual bearing noise
Candidate faults	Pass: none. Fail: F02 (bearing damage during the starved period), F13
Evidence supporting each candidate	Fail: lines and dl/dV 4.8 A per mm/s
Evidence against alternatives	Fail: F01 no longer applies (pressure restored)
Diagnostic reasoning	R-32 for both variants; R-18 for the fail variant
Final diagnosis	Pass: return to service. Fail: F02 BEARING_WEAR (secondary damage)
Recommended inspection	Fail: T2 and T3 planning
Corrective action	Pass: return to service. Fail: plan bearing replacement
Expected post-maintenance telemetry	Pass: values as the baseline
Verification	P60 4.35 to 4.65, V60 1.20 to 1.45, HB300 at most 0.06, no L1 or above for 60 minutes
Basis	[P] acceptance after the work; fail variant [I]

5.3 C17 Slow clog: drift rules lead, limits come late (600 s ramp)

Table 21. C17: case record

Element	Description
Case ID	C17
Machine state	Running; data valid; ramp 600 s (the longest allowed) so changes are one fifth of the default speed
Sensor readings and pattern	Slopes of the 60 s means: P60 -0.20 bar/min, V60 +0.37 mm/s/min, I60 +0.28 A/min [D]; HB300 rises gradually from 0.03 mm/s
Baseline	P60 4.50 bar, V60 1.31 mm/s
Threshold comparison	V60 reaches Watch (1.50) at about 30 s, V60 2.5 at about 198 s, P60 3.6 bar at about 266 s, alarm limits at about 444 s [D]
Symptoms	None for several minutes; a low pressure reading later
Candidate faults	F01 early, F02 early, F12
Evidence supporting each candidate	F01: both P60 and V60 move in the same time order; delta V / delta P about -1.8 mm/s per bar
Evidence against alternatives	F02 rejected (pressure falls); F12 rejected (V60 and HB300 follow)
Diagnostic reasoning	R-15 at 30 s, R-16 at 60 s, R-27 (P60 falls 0.3 bar in 10 minutes, here after about 90 s), R-29 estimates time to alarm from the slope
Final diagnosis	F01 CLOGGED_FILTER early, Medium confidence at first (score reaches High as P60 and V60 grow)
Recommended inspection	T1 at the next planned stop (L1 to L2)
Corrective action	Replace the cartridge at the next stop; the slow rate gives time to plan
Expected post-maintenance telemetry	As C01
Verification	As C01
Basis	[P][D]: times are r x 600 s

5.4 C18 Bearing wear: the threshold alarm baseline never fires

Table 22. C18: case record

Element	Description
Case ID	C18
Machine state	Running; data valid; ramp 120 s
Sensor readings and pattern	V60 rises from 1.31 mm/s toward 2.89 mm/s; P60 4.50 bar; I60 rises toward 102.3 A; lines grow [P][D]
Baseline	V60 1.31 mm/s
Threshold comparison	Feature rule: V60 Watch at about 14 s (r about 0.12). Platform: first samples at or over 2.5 mm/s at about 72 s (r 0.6), over it most of the time from about 96 s (r 0.8), alarm never
Symptoms	Gradual noise
Candidate faults	F02, F10, F11
Evidence supporting each candidate	F02: lines, P normal, dl/dV about 4.8
Evidence against alternatives	F01 rejected (P normal)
Diagnostic reasoning	R-15 at 14 s, R-21, R-18 once the lines pass 4 x healthy
Final diagnosis	F02 BEARING_WEAR, High confidence once lines are present
Recommended inspection	T2 at the next planned stop
Corrective action	Plan replacement; the feature rule gives about 58 s of lead at the default ramp (about 288 s at 600 s) over the first platform warning [D]
Expected post-maintenance telemetry	As C03
Verification	Section 14
Basis	[P][D]; planned evaluation measure: detection lead time against a threshold-alarm baseline, which never fires for bearing wear [P]

6. Comparison Tables and Diagnostic Evidence

6.1 Telemetry signatures across the cases

Table 23. Telemetry signature comparison

Case	P60 (bar)	V60 (mm/s)	I60 (A)	HB300 / spectrum	Level
C01 (t = 60 s)	3.49	3.15	96.2	0.447, flat	L2 to L3
C02 (45 s, 60 s ramp)	2.98	4.07	96.9	about 0.6, flat	L3 to L4
C03	4.50	1.94	97.8	low; lines	L1
C04	4.50	2.89	102.3	low; strong lines	L2
C05	4.50	2.42	100.1	low; lines	L1 to L2
C06	2.47	4.98	97.6	0.749, flat	L3 to L4
C07	3.9	3.2	not stated	0.35; lines + floor	L2
C08	3.20 flat	1.30	94.8	0.03	LS
C09	4.50	8 (spikes)	94.8	spikes	LS
C10	4.50	1.35	108	0.03	L2 (lpk)
C11	4.50	1.31	94.8	0.03	L0
C12	5.4 rising	1.6	94.8	not stated	Other
C13	3.2	1.4	94.8	0.03	L2 raw
C15A	3.63	2.89	96.0	0.4, flat	L2
C15B	4.50	2.89	102.3	low; lines	L2

6.2 Threshold interpretation

Table 24. Threshold interpretation per case

Case	Platform per-sample status	Window-based level (Document 01)	Why they differ
C01	Warn flickers from about 30 s, then warn, then alarm	L1 at 6 s, L2 at 40 s, L3 at 89 s	Window rules lead the first warn sample by about 24 s [D]
C03	Normal (no sample reaches 2.5 mm/s)	L1 (V60 above 1.50)	Feature rule catches it; platform silent [P]
C04	Warn on every sample, never alarm	L2	Bearing wear never reaches 4.0 mm/s [P]
C05	Alternating normal and warn	L1 to L2	Window mean and PF60 stop the flicker
C11	Normal	L0	Spike is not persistent
C08	Alarm or warn (per sample)	LS	Signal implausible
C09	Alarm at spikes	LS	RMS disagrees with spectrum

6.3 Detection lead of feature rules over the platform status

Table 25. Detection lead at the default 120 s ramp [D]

Case	Feature rule and time	Platform first warn sample	Platform alarm	Lead
C01 / C17 clog (V60 Watch)	6 s (r 0.05)	About 30 s (r 0.25)	About 89 s (r 0.74)	About 24 s over warn; about 83 s over alarm
C01 clog (P60 Watch 4.30 bar)	12 s (r 0.10)	About 30 s	About 89 s	About 18 s over warn
C03 / C18 bearing (V60 Watch)	14 s (r 0.12)	About 72 s (r 0.6)	Never	About 58 s over warn; alarm never fires

6.4 Sensitivity of the cases to the ramp duration

Table 26. Time to key events for ramp durations [D]

Event (r)	30 s ramp	60 s ramp	120 s ramp	300 s ramp	600 s ramp
Clog V60 Watch (0.05)	2 s	3 s	6 s	15 s	30 s
Clog first warn samples (0.25)	8 s	15 s	30 s	75 s	150 s
Clog P60 3.6 bar (0.44)	13 s	26 s	53 s	132 s	264 s
Clog alarm (0.74)	22 s	44 s	89 s	222 s	444 s
Bearing Watch (0.12)	4 s	7 s	14 s	36 s	72 s
Bearing first warn samples (0.6)	18 s	36 s	72 s	180 s	360 s

The demo-script timeline for the 60 s clog (vibration warn about 20 s, pressure under 3.6 bar about 27 s, alarms about 45 s) agrees with these values [P].

6.5 Expected root-cause outcomes for the project cases

Table 27. Expected confusion matrix of the procedure on the project cases (planned evaluation measure) [P] planned / [D]

True scenario (hidden)	Diagnosed F01	Diagnosed F02	Diagnosed other or unresolved	Cases
CLOGGED_FILTER	C01, C02, C06, C15A, C17	0	0	5 of 5 expected F01
BEARING_WEAR	0	C03, C04, C05, C15B, C18	0	5 of 5 expected F02
NORMAL	0	0	C11 (L0)	No false diagnosis

The matrix is an expected outcome of the documented rules, not a measured result: the repository lists root-cause accuracy as a planned evaluation measure and the agent is not built [P].

6.6 Candidate comparison with scores

Table 28. Evidence-weight scores (columns F01 / F02 / F03-04 / F09 / F12 / F13 / F16)

Case	Evidence matched	Scores	Top	Second	Marg in	Decision	Confiden ce
C01	E1, E3, E4, E5n, E7	11 / -8 / 6 / -3 / -4 / 0 / 3	F01 11	F03/04 6	5	clear	High
C02	E1, E3, E4, E5n, E6, E7	13 / -11 / 7 / -3 / -5 / 1 / 5	F01 13	F03/04 7	6	clear	High
C03	E2, E3, E5, E8	-6 / 11 / -5 / 1 / -3 / 0 / 3	F02 11	F16 3	8	clear	High
C04	E2, E3, E5, E8	-6 / 11 / -5 / 1 / -3 / 0 / 3	F02 11	F16 3	8	clear	High
C05	E2, E3, E5, E8	-6 / 11 / -5 / 1 / -3 / 0 / 3	F02 11	F16 3	8	clear	High
C06	E1, E3, E4, E5n, E6, E7	13 / -11 / 7 / -3 / -5 / 1 / 5	F01 13	F03/04 7	6	clear	High
C07	E1, E3, E5, E4	5 / 1 / 3 / -4 / -5 / -1 / 8	F16 8	F01 5	3	ambiguous	Medium
C08	E11, E14, E2	-8 / 2 / 0 / 1 / 7 / 1 / -8	F12 7	F02 2	5	clear	Medium
C09	E10, E12, E2	-8 / -3 / -6 / 0 / 2 / 8 / -8	F13 8	F12 2	6	clear	Medium
C10	E2, E9	-4 / 1 / -4 / 5 / 0 / 1 / -4	F09 5	F02 1	4	clear	Low
C13	E1, E14	0 / -3 / 5 / -2 / 4 / -1 / 0	F03/04 5	F12 4	1	unresolved	Low
C15A	E1, E3, E4, E5n, E7	11 / -8 / 6 / -3 / -4 / 0 / 3	F01 11	F03/04 6	5	clear	High
C15B	E2, E3, E5, E8	-6 / 11 / -5 / 1 / -3 / 0 / 3	F02 11	F16 3	8	clear	High

C11 (no persistent evidence), C12 (outside the matrix: rising pressure), C14 (data state) and C16 (verification) are not scored. Scores use the matrix of Document 03.

6.7 Diagnostic evidence

Table 29. Diagnostic evidence by fault and case

Fault	Key evidence	Cases	Rule	What rules it out
F01	P60 falling, HB300 at or above 0.24 mm/s, PMR300 below 2.5, dl/dV about 0.8	C01, C02, C06, C15A	R-06, R-17, R-25	Normal pressure; discrete lines
F02	P normal, lines at defect orders, dl/dV about 4.8, alarm never reached	C03, C04, C05, C15B	R-18, R-19, R-21, R-24	Falling pressure; flat floor
F16	Falling pressure plus lines	C07	R-20	Single-fault patterns
F12	Flat or offset pressure, no V response	C08, C13	R-03, R-34	V and HB300 following the fall
F13	RMS differs from spectrum, step spikes	C09	R-04	Lines at defect orders
F09	l _{pk} up, idle unchanged	C10	R-23	V or P change
F05	Pressure rising	C12	R-22	Falling pressure
F15	Stale, new bootId	C14	R-01, R-02	Any mechanical hypothesis

6.8 Recommended actions and verification

Table 30. Action and verification by case

Case	Inspection	Action	Verification criterion
C01	T1	Replace cartridge	P60 4.35 to 4.65; V60 1.20 to 1.45
C02	T1 then T2	Replace cartridge now; check bearing	As C01 plus spectrum
C03	T2	Plan bearing replacement	Lines absent; V60 1.3
C04	T2, T3	Replace bearings at next stop	Lines absent; V60 1.3; I60 94.8
C05	T2	Plan replacement	As C04
C06	T1, T2	Stop, replace cartridge, inspect bearings	As C01 plus 60 minute run
C07	T1 then T2	Filter first, re-check spectrum	P restored; lines absent
C08	T4	Replace transducer	Ripple and reference agree
C09	T5	Refit sensor	RMS equals spectrum
C10	T8	Correct setup	Ipk about 150 A
C11	None	Log only	No change
C12	Lines after the transducer	Clear restriction	P 4.5 bar
C13	T4 then T7	Resolve ambiguity	Depends on finding
C14	Gateway diag	Fix link	No gaps
C15	T1 or T2	Per diagnosis	Per diagnosis
C16	Section 14	Return or reopen	Baseline values

7. Lessons and Summary

7.1 Lessons from the cases

- The vibration RMS alone cannot separate the two simulated faults (C15): use the pressure and the spectrum shape.
- A threshold alarm alone misses bearing wear (C03, C04): use trend and shape [P].
- Window-based rules stop alarm flicker (C05) and rule out spikes (C11).
- Sensor and data checks come first (C08, C09, C14); a pressure alarm without a vibration response is a sensor suspect.
- When the evidence is mixed, lubrication first (C07, C13) [P].
- Always verify against the baseline after a repair (C16).

7.2 Key thresholds

Table 31. Key thresholds used in the cases

Quantity	Value	Tag
V warn / alarm / critical	2.5 / 4.0 / 7.0 mm/s	[P] / [P] / [I]
P warn / alarm / critical	3.6 / 3.0 / 2.5 bar	[P] / [P] / [I]
I warn / alarm / critical	175 / 195 / 220 A	[P] / [P] / [I]
Verification band	P60 4.35 to 4.65 bar; V60 1.20 to 1.45 mm/s	[P] / [I]
Margin rule	4 clear, 2 to 3 ambiguous, 0 to 1 unresolved	[I]

7.3 Safety and escalation notes

The cases are simulated or illustrative and are not plant history. This document is a capstone prototype, not an OEM manual. Follow the manufacturer's procedure and apply lock-out and tag-out before opening the lubrication circuit or the main drive. The agent cannot act on the machine.

7.4 References and engineering basis

- CONTEXT_PRESS_STAMP_01.md; the simulator equations and derived tables; docs/demo-script.md for the C02 timeline; the two press runbooks [P].
- Cases C07 to C14 and the fail variant of C16 are illustrative engineering examples [I].

7.5 Revision notes

Table 32. Revision history

Rev	Date	Change
1.0	2026-10-08	First issue